

GEOBARD

Turn GeoJSON into prose.

The open-source narration engine of **GaiaWM** — where map data becomes what an agent **sees**.

github.com/openfantasymap/geobard · MIT / Apache-2.0 · v0.1

Prose renders worlds.

j.r.r. tolkien · the hobbit, 1937

"In a hole in the ground there lived a hobbit. Not a nasty, dirty, wet hole... it was a hobbit-hole, and that means comfort."



peter jackson · fellowship, 2001

The round green door. The Party Tree. Smoke over terraced gardens – Hobbiton built full-scale at Matamata, NZ, and millions recognised a place they had only ever read.

- ▶ **There is no film still on this slide** — and yet you saw it.
- ▶ **Prose is the original rendering engine** — a few lines, and everyone builds the same place.
- ▶ **geobard puts that engine on tap** — data in, a place you can see out.

A scene is data around a point of view

```
{ "pov": { "lon": -79.714, "lat": 48.834,
           "heading": "NE" },
  "features": [
    { "class": "city", "name": "Neverwinter",
      "distance_m": 1400, "bearing": 47,
      "lore": "jewel of the north, rebuilt
              after the cataclysm" },
    { "class": "road", "name": "High Road",
      "distance_m": 300, "bearing": 90 },
    { "class": "river", "name": "Neverwinter River",
      "distance_m": 900, "bearing": 15 } ] }
```

anchored

everything is relative to a pov — the question is always “what surrounds you?”

standard geometry

GeoJSON from anywhere: a PostGIS sector query, SpatiaLite, a plain file.

lore rides along

non-standard fields carry world context the narrator is told to use.

Rules of perception, not a schema

the rules, verbatim from the product

- No technical or mapping language
- No coordinates or measurements
- Use visual, spatial, and sensory cues
- Distance in human terms: near, across the street, farther away, to the left...
- If unclear, stay vague rather than inventing details
- Context before description
- 1–2 short paragraphs, conversational

the voice on top

the per-world system prompt:
medieval, noir, tour-guide — how a world talks about itself.

a detail dial

detail_level turns a glance into a long look.

any model

the rules travel; the LLM behind them is interchangeable.

Data dissolves into narration

<code>"distance_m": 1400, "bearing": 47</code>	→	“off to your right, close enough to reach before dark”
<code>"lore": "jewel of the north"</code>	→	“they still call it the jewel of the north”
<code>"pov": { ... }</code>	→	the camera dissolves into “you” — no coordinate survives

“You stand where the High Road bends — it runs on just ahead.
Off to your right smoke rises over rooftops: Neverwinter,
close enough to reach before dark. They still call it the
jewel of the north. Somewhere past the fields a river talks
to itself.”

Illustrative output. The scene stays true to the data — the words are the model's, the facts are yours.

THE PROBLEM

Maps speak in coordinates.

```
{ "type": "Feature", "geometry":  
  { "type": "Point",  
    "coordinates": [-79.714, 48.834] },  
  "properties": { "class": "city",  
    "name": "Neverwinter",  
    "distance_m": 1400, "bearing": 47 } }
```



“Smoke rises from chimneys off to your right – a city, close enough to reach before the light fades. The road bends toward its gates.”

- ▶ **Agents and players don't consume bearings** — they perceive scenes.
- ▶ **Every project rebuilds this translation** — prompt spaghetti inside app code.
- ▶ **geobard extracts it** — one small service, one job, any LLM backend.

Three ways to render a place

`/narrate/window`

The window view

Describe the scene as if standing in it. Human distances, sensory cues, no map-speak. Detail level is a dial.

`/narrate/prompt`

The answer

Same scene, plus a question: “could I ford this river?” — narration becomes situated Q&A.

`/image/prompt`

The picture

Emit a prompt engineered for an image model: architecture and city structure first, from the point of view in the data.

Stateless. GeoJSON in, text out. Non-standard fields in the data become narrative context.

Same scene. Different worlds.

The **system prompt** is a per-world voice: it decides not what is there, but how a world talks about itself.

toril · medieval fantasy

"The river runs quick and cold here. Across it, torchlight – an inn, by the sound of it. Magic hangs in the evening air, faint as woodsmoke."

naples · travel companion

"You're a short walk from the water. The street to your left opens into a piazza – the espresso is worth it."

Toril's real voice ships with the world: "You describe places the way someone would when looking around. You never mention maps, coordinates, GeoJSON..." (examples above are illustrative)

Small on purpose

- ▶ **FastAPI, a handful of endpoints, no state** — nothing to babysit.
- ▶ **Speaks the OpenAI chat-completions API** — OpenRouter, Ollama, Groq, OpenAI...
- ▶ **Configuration is three env vars** — base url · api key · model.
- ▶ **Tested, CI'd, containerised** — dual-licensed MIT / Apache-2.0.

```
docker run -p 8010:8000 \  
  -e GEOBARD_OPENAI_API_KEY=sk-... \  
  -e GEOBARD_OPENAI_MODEL=meta-llama/llama-3.3-70b-instruct \  
  ghcr.io/openfantasymap/geobard:0.1
```

One call

POST /narrate/window

```
{ "geojson":      { ...the scene... },  
  "system":      ["You are the voice of a medieval world..."],  
  "detail_level": "medium" }
```

```
→ { "text": "Smoke rises from chimneys off to your right..." }
```

- ▶ **/narrate/prompt** adds a question — “What's the safest way across?”
- ▶ **/image/prompt** swaps the rules — architecture first, POV-anchored.
- ▶ **Prompt rules are the product** — human distances · sensory cues · vague beats invented.

The sensory cortex of the ghosts



- ▶ **Position-blindness is enforced by prose** — agents act on what they perceive, not on lat/lng.
- ▶ **gaia runs the same prompts in-process** — set `GAIA_GEOBARD_URL` and narration becomes a service you can scale and swap independently.

Anywhere a map needs a voice

Games & VTT

scene descriptions for NPCs, boxed text for game masters, image prompts for battle-map art.

Accessibility

describe real surroundings aloud — human terms, no jargon — from any geodata source.

Tourism & storytelling

guides that narrate a city the way a local would, tuned per neighbourhood voice.

Agent research

grounded perception for LLM agents: same world, comparable senses, swappable narrator models.

It's the engine behind EyeOnWorld — packaged for everyone else.

Where it goes next

- ▶ **Streaming** — narrate as tokens arrive — perception with latency of a heartbeat, not a page load.
- ▶ **Scene caching** — hash the (geojson, voice, detail) triple; a world mostly repeats itself.
- ▶ **Voice presets** — named, versioned culture voices — shareable between worlds and projects.
- ▶ **BYO engine per request** — callers bring their own model + key, like everything else in GaiaWM.
- ▶ **The inspector sibling** — same treatment for affordance extraction — geobard says what is there, the inspector says what you could do with it.

The world is data.

Perception is prose.

github.com/openfantasymap/geobard

gaiawm.github.io · [ghostkit](#) · [ghostdeck](#)